

English

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CLASSROOM CONTACT PROGRAMME
(Academic Session : 2024-2025)

Telegram: @NEETxNOAH

Test Pattern
NEET (UG)
MAJOR
04-04-2025

PRE-MEDICAL : ENTHUSIAST COURSE (RBT-3)

Test Booklet Code

This Booklet contains **28** pages.

E1

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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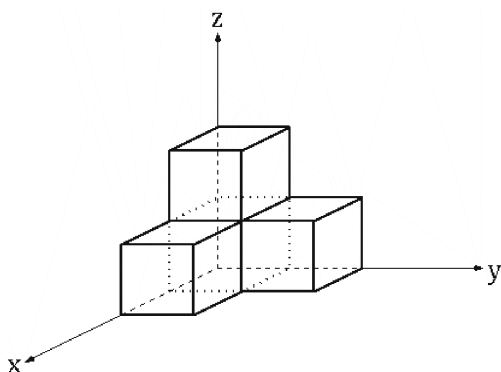
SUBJECT : PHYSICS

Topic : COM, COLLISION, MOMENTUM CONSERVATION, ROTATION MOTION

1. Two particles whose masses are 10 kg and 30 kg and their position vectors are $(\hat{i} + \hat{j} + \hat{k})$ and $(-\hat{i} - \hat{j} - \hat{k})$ respectively, would have the centre of mass at -

(1) $-\frac{(\hat{i} + \hat{j} + \hat{k})}{2}$ (2) $\frac{(\hat{i} + \hat{j} + \hat{k})}{2}$
 (3) $-\frac{(\hat{i} + \hat{j} + \hat{k})}{4}$ (4) $\frac{(\hat{i} + \hat{j} + \hat{k})}{4}$

2. Find the center of mass (x, y, z) of the following structure of four identical cubes if the length of each side of a cube is 1 unit.

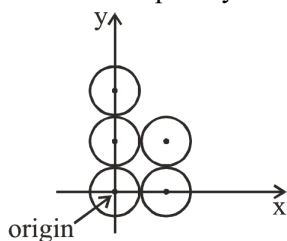


(1) $(1/2, 1/2, 1/2)$ (2) $(1/3, 1/3, 1/3)$
 (3) $(3/4, 3/4, 3/4)$ (4) $(1/2, 3/4, 1/2)$

3. A ball is dropped from rest from a height h on a smooth surface. The height attained by ball after 2nd rebound is $\frac{h}{n}$. The coefficient of restitution between ball and floor is:

(1) n^2 (2) n^{-2} (3) n^4 (4) $n^{-1/4}$

4. Five uniform circular plates, each of diameter b and mass m are laid out in a pattern shown the y-coordinate of centre of mass of the five-plate system will be :-



(1) $\frac{b}{5}$ (2) $\frac{2b}{5}$ (3) $\frac{4b}{5}$ (4) $\frac{3b}{5}$

5. **Assertion :-** If a person jumps from a height with his legs rigid, then his ankle is more injured.

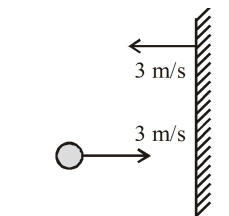
Reason :- The force will be maximum on its ankle during the collision with ground.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) (A) is false but (R) is true

6. Two semicircular rings of linear mass densities λ and 3λ and of radius R each are joining to form a complete ring. The distance of the centre of the mass of complete ring from its geometrical centre is

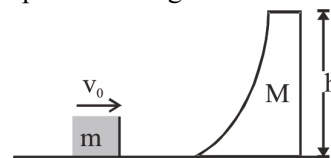
(1) $\frac{2R}{3\pi}$ (2) $\frac{R}{3\pi}$ (3) $\frac{2R}{\pi}$ (4) $\frac{R}{\pi}$

7. A highly elastic ball moving at a speed of 3 m/s approaches a wall moving towards it with a speed of 3 m/s. After the collision, the speed of the ball will be :-



- (1) 3 m/s (2) 6 m/s (3) 9 m/s (4) zero

8. A block of mass m is pushed towards the movable wedge of mass M and height h , with a velocity v_0 . All surfaces are smooth. The minimum value of v_0 for which the block will reach the top of the wedge is :

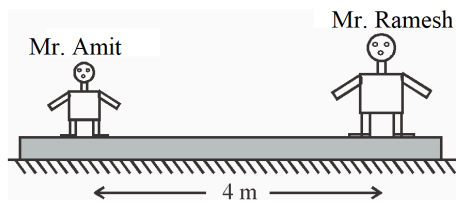


(1) $\sqrt{2gh}$ (2) $\sqrt{\frac{2ghm}{M}}$
 (3) $\sqrt{\frac{2gh(m+M)}{M}}$ (4) $\sqrt{\frac{2ghM}{(m+M)}}$

9. **Assertion :** In case of bullet fired from gun, the ratio of kinetic energy of gun and bullet is equal to ratio of mass of bullet and gun.

Reason : In firing, momentum is conserved.

- (1) Both assertion and reason are true and reason is the correct explanation of assertion.
 (2) Both assertion and reason are true and reason is not the correct explanation of assertion.
 (3) Assertion is true but reason is false.
 (4) Both assertion and reason are false.
10. Mr. Ramesh (60 kg) and Mr. Amit (50 kg) are sitting at the two extremes of a 4 m long boat (40 kg) standing still in water. They come to middle of boat to talk. Neglecting friction with water, how far does the boat moves on the water during the process ?

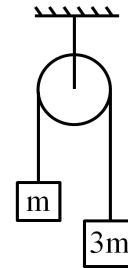


- (1) $\frac{3}{15}m$ (2) $\frac{2}{15}m$
 (3) $\frac{4}{15}m$ (4) None of these
11. The centre of mass of a solid cone along the line from the centre of the base to the vertex is at :
- (1) One-fourth of the height
 (2) One-third of the height
 (3) One-fifth of the height
 (4) None of these
12. A ball of mass 'm' moving with a speed $2v_0$ collides head on with an identical ball at rest. If 'e' is the coefficient of restitution, then what will be the ratio of velocity of two balls after collisions ?
- (1) $\frac{1-e}{1+e}$ (2) $\frac{1+e}{1-e}$
 (3) $\frac{e-1}{e+1}$ (4) $\frac{e+1}{e-1}$

13. A person of mass m is standing on one end of a plank of mass M and L and floating in water. The person moves from one end to another and stops. The displacement of the plank is:-

(1) $\frac{Lm}{(m+M)}$ (2) $Lm(M+m)$
 (3) $\frac{(M+m)}{Lm}$ (4) $\frac{LM}{(m+M)}$

14. If the system is released, then the acceleration of the centre of mass of the system :-



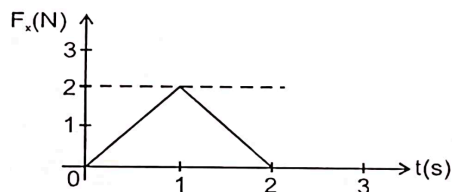
(1) $\frac{g}{4}$ (2) $\frac{g}{2}$ (3) g (4) 2g

15. **Assertion :** Kinetic energy is conserved in both, perfectly elastic & inelastic collisions.

Reason : Because both the types of collisions are identical.

- (1) Both Assertion (A) and Reason (R) are the true and Reason (R) is a correct explanation of Assertion (A).
 (2) Both Assertion (A) and Reason (R) are the true but Reason (R) is not a correct explanation of Assertion (A).
 (3) Assertion (A) is true and Reason (R) is false.
 (4) Assertion (A) and Reason (R) both are false.
16. If you know that impulse that has acted on a body of mass m, you can calculate :
- (1) its initial velocity
 (2) its final velocity
 (3) its final momentum
 (4) the change in its velocity

17. The given figure shows a plot of the time dependent force $F_x(t)$ acting on a particle in motion along the x-axis. What is the total impulse delivered by this force to the particle from time $t = 0$ to $t = 2$ second ?



- (1) 0 (2) 1 kg-m/s
(3) 2 kg-m/s (4) 3 kg-m/s
18. **Statement-I :-** If the mass of the colliding particles remains constant, then the linear velocity of the individual particles change during collision along common normal direction.
Statement-II :- A pair of equal and opposite impulses act along common normal direction.
- (1) Both statements-I and II are true and statement-II is the correct explanation of statement-I.
(2) Both statements-I and II are true but statement-II is not correct explanation of statement-I.
(3) Statement-I is true but statement-II is false.
(4) Both statements-I and II are false.
19. A solid sphere is moving and it makes an elastic collision with another stationary sphere of half of its own radius. After collision it comes to rest. The ratio of the densities of materials of second sphere and first sphere is :-
- (1) 2 (2) 4 (3) 8 (4) 16
20. A thin rod of mass m and length ℓ is oscillating about horizontal axis through its one end. Its maximum angular speed is ω when it passes through the mean position. Its centre of mass will rise upto maximum height (acceleration due to gravity is g) :-
- (1) $\frac{1}{6} \frac{\ell \omega}{g}$ (2) $\frac{1}{2} \frac{\ell^2 \omega^2}{g}$
(3) $\frac{1}{6} \frac{\ell^2 \omega^2}{g}$ (4) $\frac{1}{3} \frac{\ell^2 \omega^2}{g}$

21. A disc rolls down an inclined plane of inclination θ . If mass of disc is M , then friction force acting on disc will be :

- (1) $Mg \sin \theta$ (2) $\frac{1}{3} Mg \sin \theta$
(3) $\frac{2}{5} Mg \sin \theta$ (4) $\frac{1}{2} Mg \sin \theta$

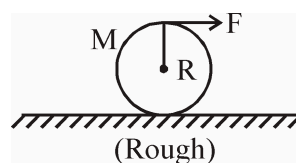
22. The radius of gyration of a rod of length L and mass M about an axis perpendicular to its length and passing through a point at a distance $L/3$ from one of its ends is :-

- (1) $\frac{\sqrt{7}}{6} L$ (2) $\frac{L}{9}$ (3) $\frac{L}{3}$ (4) $\frac{\sqrt{5}}{2} L$

23. A tap can be operated easily using two fingers because

- (1) The force available for the operation will be more
(2) This helps application of angular forces
(3) The rotational effect is caused by the couple formed
(4) The force by one finger overcomes friction and other finger provides the force for the operation

24. A hollow sphere is performing pure rolling then find acceleration of its centre of mass :-



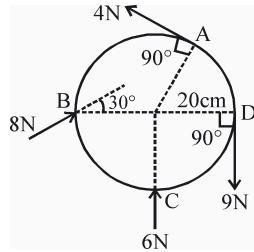
- (1) $\frac{6F}{5M}$ (2) $\frac{3F}{5M}$
(3) $\frac{2F}{3M}$ (4) None of these

25. The instantaneous angular position of a point on a rotating wheel is given by the equation :-
 $\theta = 2t^3 - 6t^2$

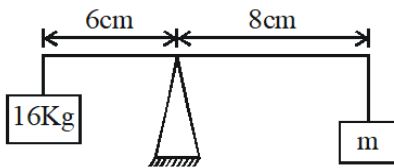
The torque on the wheel becomes zero at

- (1) $t = 0.5$ sec (2) $t = 0.25$ sec
(3) $t = 2$ sec (4) $t = 1$ sec

26. A wheel of radius 20 cm has forces applied to it as shown in the figure. The net torque produced by the forces 4 N at A, 8 N at B, 6 N at C and 9 N at D at angles indicated is :-

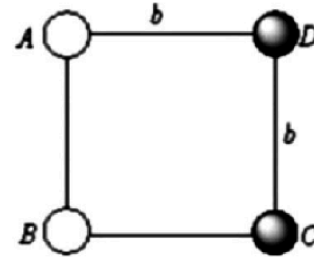


- (1) 5.4 N-m (anticlockwise)
 (2) 1.8 N-m (clockwise)
 (3) 2.0 N-m (clockwise)
 (4) 5.4 N-m (clockwise)
27. In an experiment, m mass is kept in equilibrium as shown in the picture, then tell the value of m .



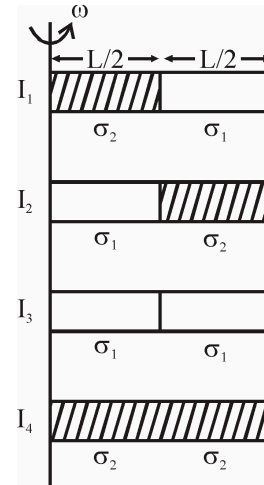
- (1) 12 Kg (2) 6 Kg (3) 8 Kg (4) 24 Kg
28. If the external forces acting on a system have zero resultant, the centre of mass :
- (1) must not move (2) must accelerate
 (3) may move (4) may accelerate
29. Consider the following statements and choose the correct option among the following.
Statement-A: In pure translational motion at any instant of time, all particles of the body have the same velocity.
Statement-B: In pure rotation of the rigid body about a fixed axis, every particle of the body moves in a circle, which lies in a plane perpendicular to the axis and has its centre on the axis.
- (1) Both A & B are correct
 (2) A is correct and B is incorrect
 (3) A is incorrect and B is correct
 (4) Neither A is correct nor B is correct

30. For spheres each of mass M and radius R are placed with their centres on the four corners A, B, C and D of a square of side b . The spheres A & B are hollow and C & D are solid. The moment of inertia of the system about side AD of square is



- (1) $\frac{8}{3}MR^2 + 2Mb^2$ (2) $\frac{8}{5}MR^2 + 2Mb^2$
 (3) $\frac{32}{15}MR^2 + 2Mb^2$ (4) $32MR^2 + 4Mb^2$

31.

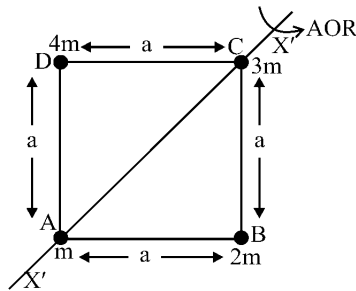


If all rods have same length and area of cross-section. Density (σ_2) > density (σ_1) then :-

- (1) $I_4 > I_2 > I_3 > I_1$ (2) $I_4 > I_3 > I_2 > I_1$
 (3) $I_1 > I_2 > I_3 > I_4$ (4) $I_4 > I_2 > I_1 > I_3$
32. A particle of mass ' m ' is projected on horizontal ground with an initial velocity of v making an angle θ with horizontal. Find out the angular momentum of particle about the point of projection when it is at its maximum height :

- (1) $\frac{mv \sin \theta \cos \theta}{2g}$ (2) $\frac{mv^3 \sin \theta \cos \theta}{2g}$
 (3) $\frac{2mv^3 \sin \theta \cos \theta}{g}$ (4) $\frac{mv^3 \sin^2 \theta \cos \theta}{2g}$

33. Find MI about XX'



- (1) ma^2 (2) $3ma^2$ (3) $6ma^2$ (4) $4ma^2$

34. The position of a particle is given by :

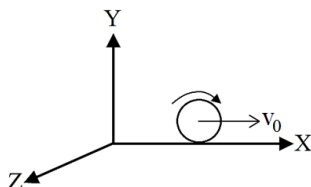
$\vec{r} = (\hat{i} + 2\hat{j} - \hat{k})$ and momentum $\vec{p} = (3\hat{i} + 4\hat{j} - 2\hat{k})$. The angular momentum is perpendicular to :-

- (1) x axis
(2) y axis
(3) z axis
(4) line at equal angles all the three axis

35. A person, with outstretched arms, is spinning on a rotating stool. He suddenly brings his arms down to his sides. Which of the following is true about system kinetic energy K and angular momentum L ?

- (1) Both K and L increase.
(2) Both K and L remain unchanged.
(3) K remains constant, L increases.
(4) K increases but L remains constant.

36. A solid sphere rolls without slipping on a rough surface and the centre of mass has a constant speed v_0 . If the mass of the sphere is m and its radius is R , then find the angular momentum of the sphere about the point of contact.



- (1) $\frac{7}{5} Mv_0R(-\hat{k})$ (2) $\frac{3}{5} Mv_0R(-\hat{k})$
(3) $\frac{Mv_0R}{2}(-\hat{k})$ (4) $\frac{2}{5} Mv_0R(-\hat{k})$

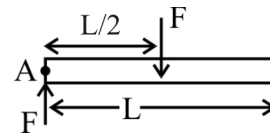
37. A solid cylinder of mass 2 kg and radius 50 cm rolls up an inclined plane of angle inclination 30° . The centre of mass of cylinder has speed of 4 m/s. The distance travelled by the cylinder on the incline surface will be : (Take $g = 10 \text{ m/s}^2$)

- (1) 2.2 m
(2) 1.6 m
(3) 1.2 m
(4) 2.4 m

38. If the angular momentum of a rotating body is increased by 200%, then its kinetic energy will be increased by –

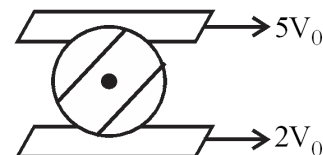
- (1) 400%
(2) 800%
(3) 200%
(4) 100%

39. Find net torque of rod about point A.



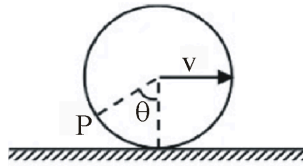
- (1) $\frac{FL}{3}$
(2) $\frac{FL}{2}$
(3) FL
(4) $2FL$

40. A cylinder of radius R is in state of rolling under shown condition find angular velocity of cylinder.

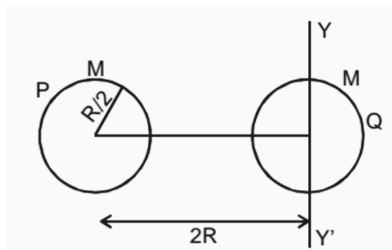


- (1) $\frac{3v_0}{2R}$ (2) $\frac{2v_0}{3R}$
(3) $\frac{3v_0}{4R}$ (4) $\frac{v_0}{2R}$

41. A hoop rolls on a horizontal ground without slipping with linear speed v . Speed of a particle P on the circumference of the hoop at angle θ is :



- (1) $2v \sin\left(\frac{\theta}{2}\right)$ (2) $v \sin\theta$
 (3) $2v \cos\left(\frac{\theta}{2}\right)$ (4) $v \cos\theta$
42. A ring of radius $2m$ and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 10 cm/s . How much work is needed to stop it ?
- (1) $3J$ (2) $30KJ$
 (3) $2J$ (4) $1J$
43. Two solid spheres each of mass M and radius $R/2$ are connected with a massless rod of length $2R$ as shown in the-figure. What will be the moment of inertia of the system about an axis passing through the centre of one of the spheres and perpendicular to the rod:

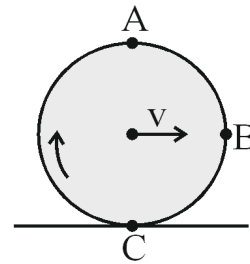


- (1) $\frac{21}{5}MR^2$ (2) $\frac{2}{5}MR^2$
 (3) $\frac{5}{2}MR^2$ (4) $\frac{5}{21}MR^2$

44. A light rod of length l has two masses m_1 and m_2 attached to its two ends. The moment of inertia of the system about an axis perpendicular to the rod and passing through the centre of mass is :

- (1) $\sqrt{m_1 m_2} l^2$
 (2) $\frac{m_1 m_2}{m_1 + m_2} l^2$
 (3) $\frac{m_1 m_2}{m_1 + m_2} l^2$
 (4) $(m_1 + m_2) l^2$

45. A solid disc rolls clockwise without slipping over a horizontal path with a constant speed v . Then the magnitude of the velocities of points A , B and C (see figure) with respect to a standing observer are, respectively :-



- (1) v, v and v
 (2) $2v, \sqrt{2}v$ and zero
 (3) $2v, 2v$ and zero
 (4) $2v, \sqrt{2}v$ and $\sqrt{2}v$

SUBJECT : CHEMISTRY

Topic : EQUILIBRIUM , REDOX REACTION

46. On heating 1 mole of PCl_5 in a closed vessel at 520 K. It dissociates as $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$. Fill in the blank in the statement : "If there is now x moles of PCl_5 , the amount of PCl_3 present would be mole(s) :-
- (1) x (2) $1 - x$
(3) $1 + x$ (4) $2x$
47. For the reaction:
 $\text{A(g)} + \text{B(g)} \rightleftharpoons 3\text{C(g)} + \text{D(g)}$
If two moles each of A and B are taken initially, then choose the true statement at equilibrium- [volume of the vessel = 1 litre, degree of dissociation = x] :-
- (1) $[\text{A}]-[\text{D}]=2-4x$ (2) $[\text{A}]-[\text{B}]=1-x$
(3) Both the above (4) None of the above
48. In each of the following equilibrium, pressure is made four times after equilibrium is set up. In which case yield of the product(s) in maximum?
- (1) $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
(2) $\text{C(s)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CO(g)} + \text{H}_2(\text{g})$
(3) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
(4) $2\text{Fe(s)} + 3\text{H}_2\text{O(g)} \rightleftharpoons \text{Fe}_2\text{O}_3(\text{s}) + 3\text{H}_2(\text{g})$
49. For the equilibrium $\text{H}_2\text{O(s)} \rightleftharpoons \text{H}_2\text{O(l)}$ more ice will formed when :-
- (1) Pressure increase
(2) Pressure decrease
(3) Temperature increase
(4) None
50. At equilibrium which is correct :-
- (1) $\Delta G = 0$ (2) $\Delta S = 0$
(3) $\Delta G = +ve$ (4) $\Delta G^\circ = 0$
51. Predict the direction of the reaction from the comparison of Q_c and K_c . Mark the incorrect statement :-
- (1) If $Q_c < K_c$, reaction goes from left to right
(2) If $Q_c = K_c$, reaction goes from left to right
(3) If $Q_c > K_c$, net reaction goes from right to left.
(4) If $Q_c = K_c$, reactants and products are at equilibrium.
52. Degree of ionisation can be affected by :-
- (1) Temperature
(2) Dilution
(3) Common ion effect
(4) All of these
53. For the reaction $\text{N}_{2(\text{g})} + 3\text{H}_{2(\text{g})} \rightleftharpoons 2\text{NH}_{3(\text{g})}$, the correct expression of equilibrium constant (K) is
- (1) $K = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$ (2) $K = \frac{[\text{N}_2][\text{H}_2]^3}{[\text{NH}_3]^2}$
(3) $K = \frac{2[\text{NH}_3]}{[\text{N}_2] \times 3[\text{H}_2]}$ (4) $K = \frac{[\text{N}_2] \times 3[\text{H}_2]}{2[\text{NH}_3]}$
54. The equilibrium constant for the reaction $\text{SO}_{3(\text{g})} \rightleftharpoons \text{SO}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})}$ is $K_C = 4.9 \times 10^{-2}$. The value of K_C for the reaction $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ will be
- (1) 416 (2) 2.40×10^{-2}
(3) 9.8×10^{-2} (4) 4.9×10^{-2}
55. For the fazeous reaction
 $\text{PCl}_5 \rightleftharpoons \text{PCl}_3 + \text{Cl}_2$
(g) (g) (g)
Find the $\frac{K_C}{K_P}$ value of
- (1) $(RT)^1$ (2) $(RT)^{-1}$
(3) $(RT)^{-2}$ (4) $(RT)^2$

56. At a certain temperature, the following reactions have the equilibrium constants as shown below :
- $$\text{S(s)} + \text{O}_2(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}); K_1 = 5 \times 10^{52}$$
- $$2 \text{S(s)} + 3 \text{O}_2(\text{g}) \rightleftharpoons 2 \text{SO}_3(\text{g}); K_2 = 10^{29}$$
- What is the equilibrium constant K_3 for the reaction at the same temperature ?
- $$2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{SO}_3(\text{g})$$
- (1) 2.5×10^{76} (2) 4×10^{23}
 (3) 4×10^{-77} (4) None of these
57. For an endothermic reaction equilibrium constant at T_1 and T_2 are respectively K_1 and K_2 . If $K_1 > K_2$ then :-
- (1) $T_1 > T_2$ (2) $T_1 < T_2$
 (3) $T_1 = T_2$ (4) None
58. If some He gas is introduced into the equilibrium $\text{PCl}_{5(\text{g})} \rightleftharpoons \text{PCl}_{3(\text{g})} + \text{Cl}_{2(\text{g})}$ at constant pressure and temperature then equilibrium constant of reaction:
- (1) Increase (2) Decrease
 (3) Unchange (4) Nothing can be said
59. At certain temperature compound AB_2 dissociates according to reaction $2\text{AB}_2(\text{g}) \rightleftharpoons 2\text{AB}(\text{g}) + \text{B}_2(\text{g})$. If α is small as compared to unity. Which reaction is current for K_p , α & initial pressure ?
- (1) $\frac{P\alpha^3}{2}$ (2) $\frac{P\alpha^2}{3}$
 (3) $\frac{P\alpha^3}{3}$ (4) $\frac{P\alpha^2}{2}$
60. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on the addition of graphite. If the total pressure at equilibrium is 0.9 atm, the value of K_p is :-
- (1) 6.4 atm (2) 0.18 atm
 (3) 1.8 atm (4) 3 atm
61. For pure water sum of concentration of H^+ & OH^- ion at 90°C is-
- (1) 10^{-14} M (2) $2 \times 10^{-7} \text{ M}$
 (3) $2 \times 10^{-6} \text{ M}$ (4) 10^{-12} M
62. What is the resultant pH of solution if $\text{pH} = 3$ and $\text{pH} = 5$ same volume solution mix together :-
- (1) 3.5 (2) 4 (3) 3.3 (4) 3.8
63. K_b for NH_4OH is 1.8×10^{-5} , what is the $[\text{OH}^-]$ in its solution of 0.1 M concentration :-
- (1) $1.8 \times 10^{-6} \text{ M}$ (2) $1.34 \times 10^{-3} \text{ M}$
 (3) $5 \times 10^{-4} \text{ M}$ (4) $4.2 \times 10^{-3} \text{ M}$
64. Calculate the molar solubility of $\text{Ni}(\text{OH})_2$ in 0.10 M NaOH. The solubility product of $\text{Ni}(\text{OH})_2$ is 2×10^{-15} .
- (1) $2 \times 10^{-13} \text{ M}$ (2) $3.2 \times 10^{-8} \text{ M}$
 (3) $5 \times 10^{-5} \text{ M}$ (4) $3 \times 10^{-17} \text{ M}$
65. The ionisation constant of a tribasic acid is K_a . If its first, second and third ionisation constant are K_{a_1} , K_{a_2} and K_{a_3} respectively then:
- (1) $K_a = K_{a_1} \times K_{a_2} \times K_{a_3}$
 (2) $K_a = \frac{K_{a_1}}{K_{a_2} \times K_{a_3}}$
 (3) $K_{a_2} = \frac{K_{a_1} \times K_a}{K_{a_3}}$
 (4) None of these
66. pH of human blood is 7.4. Then H^+ concentration will be:
- (1) 4×10^{-8} (2) 2×10^{-8}
 (3) 4×10^{-4} (4) 2×10^{-4}
67. Consider various species generated when H_3PO_4 dissolved in water. Among these, the conjugate acid of HPO_4^{2-} is :
- (1) H_3PO_4 (2) H_2PO_4^-
 (3) PO_4^{3-} (4) H_3O^+

68. Which of the following could act as a Lewis acid but not Bronsted acid :-
 (1) HCl (2) H₂SO₄
 (3) HSO₃⁻ (4) SO₃
69. An acid solution of pH = 6 is diluted 100 times. The pH of solution becomes
 (1) 6.98 (2) 6 (3) 4 (4) 8
70. Out of the following, amphiprotic species are :-
 (I) H₂PO₄⁻ (II) HPO₄²⁻
 (III) HCO₃⁻ (IV) CH₃CO₂⁻
 (V) HPO₄²⁻
 (1) (I), (II), (III), (IV) (2) (I), (IV)
 (3) (III), (V) (4) (II), (III), (V)
71. Which of the following solutions will have pH close to 2.0 -
 (1) 50 ml of $\frac{M}{10}$ HCl + 50 ml. of $\frac{M}{10}$ KOH
 (2) 55 ml of $\frac{M}{10}$ HCl + 45 ml of $\frac{M}{10}$ KOH
 (3) 10 ml of $\frac{M}{10}$ HCl + 40 ml of $\frac{M}{10}$ KOH
 (4) 75 ml of $\frac{M}{10}$ HCl + 25 ml. of $\frac{M}{10}$ KOH
72. What is the pOH - pK_b for the buffer solution of aqueous solution of 0.20M NH₃ and 0.40M NH₄Cl if K_b for ammonia = 10⁻⁵ -
 (1) 0.50 (2) 0.60 (3) 0.30 (4) 0.75
73. Which of the following salts is neutral in water ?
 (1) KCl (2) NH₄NO₃
 (3) NH₄CN (4) NH₄OH
74. The correct order of solubility for the given salts CuCl (K_{sp}=1.7×10⁻⁶), PbCO₃ (K_{sp}= 7.4 × 10⁻¹⁴) and SnS (K_{sp}=1.0 × 10⁻²⁵) is :
 (1) CuCl > SnS > PbCO₃
 (2) CuCl > PbCO₃ > SnS
 (3) PbCO₃ > SnS > CuCl
 (4) SnS > PbCO₃ > CuCl
75. Which of the following combinations will make a buffer solution ?
 (i) CH₃COONa (3 mol) + HCl (2 mol)
 (ii) CH₃COOH (2 mol) + NaOH (3 mol)
 (iii) CH₃COOH (3 mol) + CH₃COONa (3 mol)
 (iv) NH₄OH (3 mol) + HCl (2 mol)
 (1) (i), (ii) (2) (i), (iii), (iv)
 (3) (ii), (iii) (4) (i), (ii), (iii), (iv)
76. **Assertion** : pH of a neutral solution is always 7 at 25°C.
Reason : pH of a solution does not depend on temperature.
 (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
 (2) **Assertion** is true but **Reason** is false.
 (3) **Assertion** is false but **Reason** is true.
 (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
77. On which of the following Ostwald dilution law is applicable ?
 (1) CH₃COONa (2) CH₃COOH
 (3) HCl (4) NaOH
78. K_{sp} of Ag₂CrO₄ is y, then calculate value of [Ag⁺] in saturated solution ?
 (1) $\left(\frac{y}{2}\right)^{\frac{1}{3}}$ (2) $\frac{1}{(2y)^{\frac{1}{3}}}$
 (3) $\left(\frac{y}{4}\right)^{\frac{1}{3}}$ (4) $\left(\frac{y}{4}\right)^{\frac{1}{2}}$
79. An aqueous solution of hydrogen sulphide shows the equilibrium; H₂S ⇌ H⁺ + HS⁻
 If dilute hydrochloric acid is added to an aqueous solution of H₂S, without any change in temperature :-
 (1) The equilibrium constant will change
 (2) The concentration HS⁻ will increase
 (3) The concentration of undissociated hydrogen sulphide will decrease.
 (4) The concentration of HS⁻ will decrease

80. Following reaction describes the rusting of iron
 $4\text{Fe} + 3\text{O}_2 \rightarrow 4\text{Fe}^{3+} + 6\text{O}^{2-}$
 Which one of the following statement is incorrect?
 (1) This is an example of redox reaction
 (2) Metallic iron is reduced to Fe^{3+}
 (3) Oxygen is an oxidising agent
 (4) Metallic iron is a reducing agent
81. The equivalent weight of divalent metal is W. The molecular weight of its chloride is :
 (1) $W + 35.5$ (2) $W + 71$
 (3) $2W + 71$ (4) $2W + 35.5$
82. 1.520 g of the metal hydroxide of a metal on ignition gave 0.995 g of oxide. The equivalent mass of metal is :
 (1) 1.520 (2) 0.995
 (3) 19.00 (4) 9.00
83. In the reaction :
 $2\text{CuSO}_4 + 4\text{KI} \rightarrow \text{Cu}_2\text{I}_2 + 2\text{K}_2\text{SO}_4 + \text{I}_2$
 The ratio of equivalent mass of CuSO_4 to molecular mass :-
 (1) $\frac{1}{8}$ (2) $\frac{1}{4}$
 (3) $\frac{1}{2}$ (4) 1
84. For the redox reaction,
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
 the correct coefficients of the reactants for the balanced reaction are respectively
 MnO_4^- , $\text{C}_2\text{O}_4^{2-}$, H^+ :
 (1) 2, 5, 16 (2) 16, 3, 12
 (3) 15, 16, 12 (4) 2, 16, 5
85. Which of following is reductant :
 (1) $\text{K}_2\text{Cr}_2\text{O}_7$ (2) KMnO_4
 (3) H_2SO_4 (4) H_2S
86. In which of the following compounds iron has lowest oxidation state:-
 (1) $\text{FeSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
 (2) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (3) $\text{Fe}(\text{CO})_5$
 (4) $\text{Fe}_{0.94}\text{O}$
87. The oxidation number of iodine in HIO_4 , H_3IO_5 and H_5IO_6 are, respectively :
 (1) +1, +3, +7 (2) +7, +7, +3
 (3) +7, +7, +7 (4) +7, +5, +3
88. Which of the following is disproportionation reaction ?
 (1) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
 (2) $\text{CH}_4 + 4\text{Cl}_2 \rightarrow \text{CCl}_4 + 4\text{HCl}$
 (3) $2\text{F}_2 + 2\text{OH}^- \rightarrow 2\text{F}^- + \text{OF}_2 + \text{H}_2\text{O}$
 (4) $2\text{NO}_2 + 2\text{OH}^- \rightarrow \text{NO}_2^- + \text{NO}_3^- + \text{H}_2\text{O}$
89. Equivalent mass of oxidizing agent in the reaction is
 $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 3\text{S} + 2\text{H}_2\text{O}$ is :
 (1) 8 (2) 16 (3) 32 (4) 64
90. A compound containing x, y and z atoms oxidation no of x is +3, y is -5 and z is +1 then the possible formula of compound is :-
 (1) xyz (2) xyz_2
 (3) x_2yz (4) $(\text{xy})_2\text{z}$

SUBJECT : BIOLOGY

Topic : REPRODUCTION

91. Read the following statements and select the correct option

Statement A: All copulations do not lead to fertilization and pregnancy.

Statement B: Fertilization is possible only if ovum and sperms are transported simultaneously to the ampullary region.

- (1) A is true and B is incorrect statement
(2) Both A and B statements are correct
(3) Only B statement is correct
(4) Both A and B are incorrect

92. How many structures in the list given below are haploid ?

Spermatid, secondary oocyte, primary spermatocyte, ovum, sperm, oogonia, spermatogonia, polar body

- (1) Six (2) Four (3) Two (4) Five

93. These are some events associated with reproduction and embryonic development. Arrange these from first to last and select the **correct** answer.

- (a) Starts the formation of corpus luteum
(b) Cleavage
(c) Fertilisation
(d) Ovulation
(e) Implantation
(f) Formation of Graafian follicle
(g) Blastocyst
(h) Morula
(i) Zygote

- (1) f, d, a, c, i, b, h, g, e
(2) f, d, i, c, a, b, h, g, e
(3) a, f, d, c, b, i, h, g, e
(4) i, h, g, f, d, c, b, e, a

94. Menstruation is initiated by

- (1) A sudden release of FSH from the anterior pituitary
(2) A lack of estrogen and progesterone due to degeneration of the corpus luteum
(3) An increased release of estrogen and progesterone from the corpus luteum
(4) Sudden drop in FSH

95. Match the following :-

	Method		Action
(A)	Hysterectomy	(a)	Surgical removal of uterus
(B)	Tubectomy	(b)	Removal of ovaries
(C)	Oophorectomy	(c)	Cutting and ligating fallopian tube
(D)	Vasectomy	(d)	Cutting and tying vas deferens.

- (1) A-d, B-c, C-b, D-a
(2) A-c, B-a, C-b, D-d
(3) A-b, B-c, C-a, D-d
(4) A-a, B-c, C-b, D-d

96. Relaxin hormone is secreted by.

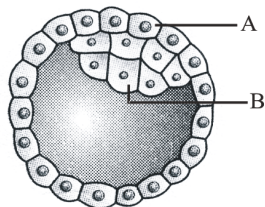
- (A) Placenta
(B) Sec. follicle
(C) Graafian follicle
(D) Corpus luteum

- (1) A and B (2) A and D
(3) B and C (4) B and D

97. In parturition process which of the following does not happen :-

- (1) Oxytocin hormone is secreted by posterior Pituitary.
(2) Relaxin hormone responsible for narrowing of pelvic cavity.
(3) Progesterone hormone secretion is stopped.
(4) General position of foetus is occipitoanterior.

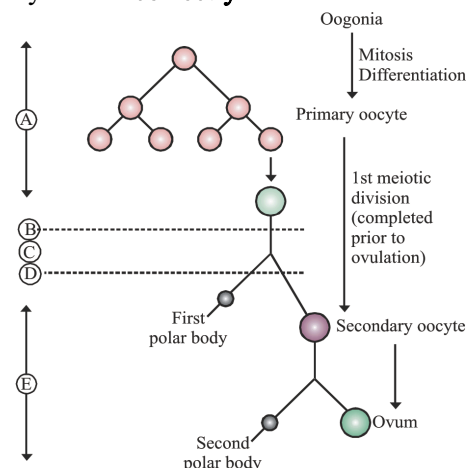
98. At the time of parturition cortisol hormone release from
- (1) Corpus luteum
 - (2) Placenta
 - (3) Adrenal gland of foetus
 - (4) Pituitary gland of mother
99. **Assertion :-** During fertilisation, only one sperm enters the ovum.
Reason :- Entry of 1st sperm induce the change in zona pellucida.
- (1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
 - (2) Assertion is true but Reason is false.
 - (3) Assertion is false but Reason is true.
 - (4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
100. When released from ovary, human egg contains:-
- (1) X/Y chromosome
 - (2) 2 X chromosome
 - (3) 1 X chromosome
 - (4) XY chromosome
101. Identify A and B and their respective functions:



Blastocyst

	A	B	Function of	
			A	B
(1)	Trophoblast	Inner cell mass	Attach to the endometrium	Differentiated as embryo
(2)	Trophoblast	Inner cell mass	Differentiated as embryo	Attach to the Endometrium
(3)	Mesoderm	Inner cell mass	Differentiated as embryo	Attach to the Endometrium
(4)	Ectoderm	Mesoderm	Attach to the endometrium	Differentiated as embryo

102. All the following events occurs during natural menopause except one :-
- (1) Progesterone levels in blood decreases
 - (2) FSH and LH levels in blood decreases
 - (3) Estrogen levels in blood decreases
 - (4) Uterine changes are stopped
103. Interphase of cleavage division consists of
- (1) only G₁ phase
 - (2) only S phase
 - (3) only G₂ phase
 - (4) all G₁, S and G₂ phases
104. **Assertion (A) :-** Corpus luteum is essential for maintenance of Endometrium.
Reason (R) :- Corpus luteum releases progesterone.
- (1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
 - (2) Assertion is true but Reason is false.
 - (3) Assertion is false but Reason is true.
 - (4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
105. The following refers to schematic representation of oogenesis. Identify A to E **correctly**.



- (1) (A) Foetal life (B) Birth (C) Puberty (D) Adult reproductive life (E) Childhood
- (2) (A) Foetal life (B) Birth (C) Childhood (D) Puberty (E) Adult reproductive life
- (3) (A) Adult reproductive life (B) Birth (C) Puberty (D) Childhood (E) Foetal life
- (4) (A) Birth (B) Childhood (C) Foetal life (D) Puberty (E) Adult reproductive life

106. Storage of sperms takes place in :-

- (1) Epididymis
- (2) Ampulla of vas deferens
- (3) Seminal vesicle
- (4) Both (1) and (2)

107. Choose the **incorrect** regarding decapacitation :-

- (1) Occurs in vas deferens before fertilization
- (2) Sperms motility is suppressed
- (3) Thin layering (of cholesterol) on apex/acrosome of sperm
- (4) Occurs in epididymis before fertilization

108. Which accessory sex gland in human being is unpaired ?

- (1) Bartholin's gland
- (2) Seminal vesicle
- (3) Cowper gland
- (4) Prostate gland

109. Liberation of sperms from sertoli cells is called :-

- (1) Semination
- (2) Spermiation
- (3) Ejaculation
- (4) Insemination

110. Match the following and choose the **correct** answer :-

- | | |
|----------------------|---------------------|
| (A) Implantation | (i) Vagina |
| (B) Capacitation | (ii) Ovary |
| (C) Folliculogenesis | (iii) Uterus |
| (D) Fertilisation | (iv) Fallopian tube |

- (1) A-(iv), B-(i), C-(iii), D-(ii)
- (2) A-(iii), B-(ii), C-(iv), D-(i)
- (3) A-(iii), B-(i), C-(ii), D-(iv)
- (4) A-(iii), B-(iv), C-(ii), D-(i)

111. Degeneration of corpus luteum occurs due to inhibition of secretion of hormone :-

- (1) FSH
- (2) LH
- (3) Estrogen
- (4) Progesterone

112. Which of the following hormone responsible for milk formation in Human female ?

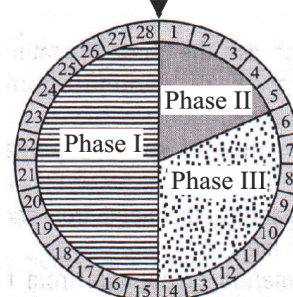
- (1) Prolactin
- (2) Oxytocin
- (3) Estrogen
- (4) Progesterone

113. Generally fertilisation takes place in :-

- (1) Isthmus
- (2) Ampulla
- (3) Infundibulum
- (4) Uterus

114. Refer the given schematic representation of menstrual cycle showing three phases I, II and III. Select the option that correctly matches these phases with the following events :-

Cycle begins here ↓

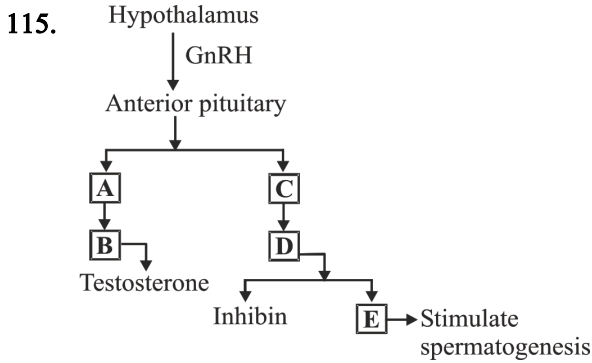


(i) FSH secreted by the anterior pituitary stimulates the ovarian follicle to secrete oestrogen which stimulates the proliferation of endometrium.

(ii) LH secreted by the anterior pituitary stimulates the development of corpus luteum which secretes progesterone

(iii) Reduced production of LH from anterior pituitary causes degeneration of corpus luteum, thereby menstrual flow starts.

- | | | |
|------------------------|---------------------|----------------------|
| (1) Phase - I
(i) | Phase - II
(ii) | Phase - III
(iii) |
| (2) Phase - I
(iii) | Phase - II
(i) | Phase - III
(ii) |
| (3) Phase - I
(ii) | Phase - II
(iii) | Phase - III
(i) |
| (4) Phase - I
(iii) | Phase - II
(ii) | Phase - III
(i) |



Find out A, B, C, D and E respectively :-

	A	B	C	D	E
(1)	Testis	Leyding cells	Sertoli cells	AMH	ABP
(2)	ICSH	Testis	AMH	Sertoli cells	ABP
(3)	LH	Interstitial cells	FSH	Sertoli cells	AMH
(4)	ICSH	Leyding cells	FSH	Sertoli cells	ABP

116. Optimal temperature for spermatogenesis is :-

- (1) Normal body temperature (37°C)
- (2) $43^{\circ} - 45^{\circ}\text{C}$
- (3) $39^{\circ} - 40^{\circ}\text{C}$
- (4) $34.5 - 35^{\circ}\text{C}$

117. During oogenesis, the meiosis division in secondary oocyte is arrested at which stage ?

- (1) Prophase I (2) Prophase II
- (3) Metaphase I (4) Metaphase II

118. How many sperm and ova will produce from 40 spermatogonium and 40 oogonium respectively.

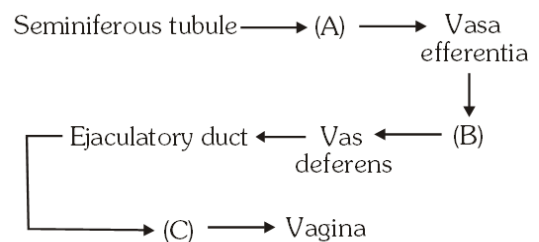
- (1) 160 sperm and 160 ova
- (2) 40 sperm and 140 ova
- (3) 160 sperm and 40 ova
- (4) 40 sperm and 40 ova

119. **Assertion:** During pregnancy the levels of hormones like estrogen, progesterone, cortisol, prolactin, thyroxine etc are increased several folds in the maternal blood.

Reason: Increased production of these hormones is essential for supporting the fetal growth, metabolic changes in the mother and maintenance of pregnancy.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

120. Here path of sperms in mammals is shown :-



	A	B	C
(1)	Epididymis	Testes	Urethra
(2)	Rete testis	Epididymis	Urethra
(3)	Ovary	Epididymis	Uterus
(4)	Testes	Rete testis	Epididymis

121. Thickness of uterine wall maximum will be, on which day of menstrual cycle approximately ?

- (1) 14th day (2) 10th day
- (3) 28th day (4) 23th day

122. In Hysterectomy which of the following condition will occurs :-

- (1) Ovarian changes present, fertilisation absent
- (2) Ovarian changes absent, fertilisation present
- (3) Both ovarian changes and fertilisation absent
- (4) Both ovarian changes and fertilisation present

123. Match the Column-A with Column-B :-

Column-A		Column-B	
(A)	Follicular phase	(i)	Release of secondary oocyte
(B)	Luteal phase	(ii)	Development of follicle
(C)	Menstrual phase	(iii)	Formation of corpus luteum
(D)	Ovulatory phase	(iv)	Breakdown of endometrium

- (1) A-(i), B-(ii), C-(iv), D-(iii)
 (2) A-(ii), B-(iii), C-(i), D-(iv)
 (3) A-(ii), B-(iii), C-(iv), D-(i)
 (4) A-(iv), B-(ii), C-(i), D-(iii)

124. Given below are two statements :

Statement I : Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.

Statement II : The cavity of the cervix is called cervical canal which along with vagina forms birth canal.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are false.
 (2) **Statement I** is correct but **Statement II** is false.
 (3) **Statement I** incorrect but **Statement II** is true.
 (4) Both **Statement I** and **Statement II** are true.

125. Which one of the following is not the function of placenta ?

- (1) Facilitates removal of CO₂ and waste material from embryo
 (2) Secrete oxytocin during parturition
 (3) Secrete estrogen & progesterone
 (4) Faciliates supply of oxygen and nutrients to embryo

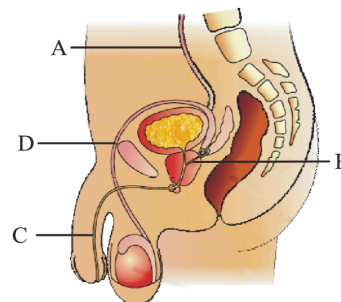
126. Which one of the following is not a function of sertoli cell.

- (1) Phagocyte the dead or injured sperm
 (2) Release AMH, ABP and inhibin
 (3) Formation of blood testis 'Barrier'
 (4) Release the testosterone

127. In males at the end of first meiotic division, which structure forms :-

- (1) Primary spermatocyte
 (2) Secondary spermatocyte
 (3) Spermatids
 (4) Spermatogonia

128. It is a diagrammatic sectional view of male reproductive system, In which identify common path for the semen and urine :



- (1) A (2) B (3) D (4) C

129. Match the following column and select the correct option -

Column-I		Column-II	
(a)	Head	(i)	Enzyme
(b)	Middle piece	(ii)	Sperm motility
(c)	Acrosome	(iii)	energy chamber
(d)	Tail	(iv)	Chromatin material

	[a]	[b]	[c]	[d]
1.	(i)	(iii)	(ii)	(iv)
2.	(iv)	(i)	(iv)	(ii)
3.	(iv)	(iii)	(i)	(ii)
4.	(i)	(ii)	(iii)	(iv)

130. Given below are two statements :

Statement-I : Gonadotropins increases gradually during the follicular phase of menstrual cycle.

Statement-II : Low level of progesterone are main cause of shedding endometrium during menstrual cycle.

- (1) Both statement-I and statement-II are false
- (2) Statement-I is true but statement-II is false
- (3) Statement-I is false but statement-II is true
- (4) Both statement-I and statement-II are true

131. Match the column-I with column-II :-

	Column-I		Column-II
I	Testes	a.	10-12 cm long
II	Ovary	b.	4-5 cm long
III	Fallopian tube	c.	2-4 cm long

- (1) I-a, II-b, III-c
- (2) I-b, II-c, III-a
- (3) I-c, II-b, III-a
- (4) I-a, II-c, III-b

132. Choose the **correct** option for filling up the blanks:-

The human male ejaculates about__ million sperms during a coitus of which, for normal fertility, at least__ percent sperms must have normal shape and size and least__ percent of them must show vigorous motility.

- (1) 100–200,40,60 (2) 200–300,60,40
- (3) 300–400,50,30 (4) 500,70,70

133. The tertiary follicle is characterised by _____

- (1) Presence of oocyte
- (2) Presence of theca externa
- (3) Presence of theca interna
- (4) Presence of antrum

134. The region outside the seminiferous tubule which contain small blood vessels, interstitial cells and immunologically competent cells is present called:-

- (1) Perivitelline space
- (2) Peritesticular space
- (3) Interstitial space
- (4) Inguinal space

135. Match the following :

	Column-A		Column-B
A	Embryo's heart is formed.	i	End of 6th month.
B	Foetus develops limbs and digits.	ii	End of 3rd month.
C	Eye-lids separate and eyelashes are formed.	iii	End of 2nd month
D	External genital organs are well developed.	iv	After 1 month of pregnancy

- (1) (A) iv (B) iii (C) ii (D) i
- (2) (A) iv (B) iii (C) i (D) ii
- (3) (A) iv (B) i (C) ii (D) iii
- (4) (A) iii (B) iv (C) i (D) ii

136. Select the **incorrect** match from the following-

- (1) Amorphophallus - Insect pollination
- (2) Viola - Air pollination
- (3) Vallisneria - Water pollination
- (4) Grass - Wind pollination

137. Monoecious plants with unisexual flowers will prevent :

- (1) Autogamy
- (2) Geitonogamy
- (3) Both autogamy and geitonogamy
- (4) Pollination is not possible

138. Read the following four statement (A-D)

- (A) Flowers are the morphological and embryological marvels and the site of sexual reproduction.
(B) Anther is attached to proximal end of the filament of stamen.
(C) Androecium and gynoecium represent the non essential parts of flower.
(D) The number and length of stamens are variable in flowers of different species

How many of the above statements are right ?

- (1) Four (2) One (3) Two (4) Three

139. Select the **correct** statement -

- (1) If the female parent bears bisexual flowers, emasculation is done
(2) If the female parent produces unisexual flower, emasculation is done
(3) If male parent produces bisexual flowers, emasculation is done
(4) Emasculation is not required if female parent is bisexual

140. Seeds offer several advantages to angiosperms. Which of the following is not true for this-

- (1) Seeds have better adaptive strategies for dispersal to new habitat
(2) The hard seed coat provides protection to young embryo.
(3) Seeds are product of sexual reproduction, thus they generate new genetic combinations leading to variation.
(4) Seed formation is always dependent on water.

141. How many microspore mother cells are responsible for the formation of 80 microspores in Angiosperms ?

- (1) 10 (2) 20
(3) 40 (4) 80

142. What will be ploidy of the cells of nucellus, megaspore mother cell, functional megaspore and embryo sac respectively

- (1) n , $2n$, $3n$, $4n$
(2) n , $3n$, $4n$, $2n$
(3) $2n$, $2n$, n , n
(4) $3n$, $4n$, $2n$, n

143. In majority of angiosperm female gametophyte is :-

- (1) 7 celled 7 nucleated structure
(2) 7 celled 8 nucleated structure
(3) 7 celled 6 nucleated structure
(4) 4 celled 4 nucleated structure

144. Transfer of pollen grains from the anther to the stigma of another flower of same plant is called.

- (1) Xenogamy (2) Autogamy
(3) Geitonogamy (4) Chasmogamy

145. Go through the following statements :

- (I) Flowers are small. They are often packed in inflorescence
(II) Flowers are colourless, nectarless and odourless
(III) Well exposed stamens
(IV) Pollen grains are produced in large numbers which are light and non-sticky
(V) Flower often have a single ovule in each ovary
(VI) Stigma is large and often feathery

The above contrivances favour :

- (1) Cross pollination
(2) Anemophily (pollination by wind)
(3) Ornithophily (pollination by birds)
(4) Entomophily (pollination by insects)

146. Which of the following is **false** ?

- (1) Wind pollination is quite common in grasses
- (2) Hydrophily is limited to about 30 genera, mostly monocots
- (3) Both wind and water pollinated flowers are not very colourful and do not produce nectar
- (4) Mostly Angiosperm plants pollinated by water

147. Pollen grain _____ has prominent apertures called _____ where sporopollenin is _____

- (1) Intine, germ pore, present
- (2) exine, germ pore, absent
- (3) exine, germ pore, present
- (4) exine, foramen ovale, absent

148. Select the correct option :

Statement-I : In Pea plant, endosperm completely consumed by developing embryo.

Statement-II : In grass family the cotyledon is called scutellum that is situated towards one side of embryonal axis.

- (1) Statement-I and Statement-II are correct.
- (2) Statement-I and Statement-II are incorrect.
- (3) Statement-I is correct and Statement-II are incorrect.
- (4) Statement-I is incorrect and Statement-II are correct.

149. Which of the following is not an adaptation for anemophily ?

- (1) Presence of well exposed stamens
- (2) Often presence of single ovule in an ovary
- (3) Sticky pollen grain
- (4) Large and often feathery stigma

150. The coconut water from tender coconut & the surrounding white kernel is respectively an example of :-

- (1) Cellular endosperm : free nuclear endosperm
- (2) Free nuclear endosperm : cellular endosperm
- (3) Helobial endosperm : free nuclear endosperm
- (4) Free nuclear endosperm : Helobial endosperm

151. **Assertion** : Cleistogamous flowers are invariably autogamous.

Reason : These flowers do not open at all.

- (1) Both assertion & reason are true & the reason is the correct explanation of the assertion.
- (2) Both assertion & reason are true but reason is not the correct explanation of the assertion.
- (3) Assertion is true but the reason is false.
- (4) Both assertion & reason are false

152. The correct sequence of embryo development in most of the Angiosperms is :-

- (1) Mature embryo → Heart shaped → Globular
- (2) Heart shaped → Globular → Mature embryo
- (3) Globular → Heart shaped → Mature embryo
- (4) Globular → Mature embryo → Heart shaped

153. In the list given below how many of them are non-endospermic seeds -

Pea, Maize, Castor, Groundnut. Wheat, Barley.

- (1) Five (2) Two (3) Six (4) Four

154. Which of the following statements correctly describes the role of the ovary in flowering plants?

- (1) It produces pollen grains for fertilization.
- (2) It develops into a fruit after fertilization, enclosing the seeds.
- (3) It is responsible for attracting pollinators through its color.
- (4) It provides nutrients to the pollen tube during its growth.

155. Arrange the following events of sexual reproduction in **correct** sequence.

- (a) Embryogeny
- (b) Triple fusion
- (c) Microsporogenesis
- (d) Pollen pistil interaction

- (1) d, b, a, c (2) d, c, a, b
- (3) d, c, b, a (4) c, d, b, a

156. An endosperm of a flowering plant has 33 chromosomes, then the number of chromosomes would be:-

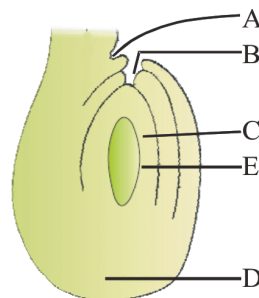
- (1) 11 in gametes (2) 22 in gametes
- (3) 44 in embryo (4) 44 in cell of stem

157. Read the following statement A-D and select **correct** statement(s).

- (A) Functional megaspore on mitotic divisions produce 8 nucleated embryosac.
- (B) 6 Nuclei are surrounded by separate cell wall and rest 2 are polar nuclei.
- (C) Double fertilization in embryosac include syngamy and triple fusion.
- (D) Formation of endosperm in angiosperm takes place before fertilization.

- (1) B, C, D (2) A, B, C
- (3) D only (4) B, C only

158.



Choose the **correct** option :-

- (1) A = Funicle B = Hilum
- (2) D = Integument E = Nucellus
- (3) E = Embryo sac A = Hilum
- (4) A = Raphe C = Mircopyle

159. If the number of chromosomes in pollen grain are 28. Then what will be the number of chromosome in nucellus:

- (1) 14 (2) 28 (3) 56 (4) 84

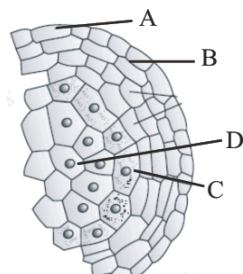
160. Which of the following statement is not **correct** for pollination ?

- (1) In monoecious plant, autogamy is prevented but not geitonogamy
- (2) In cleistogamous flower autogamy and geitonogamy both can happen
- (3) Hydrilla shows hypohydrophily
- (4) Wind pollination is quite common in grasses.

161. Select the **incorrect** statement from the following-

- (1) Most common type of female gametophyte development in angiosperm is monosporic
- (2) Each ovule has one or two protective envelopes called integuments
- (3) Small opening called micropyle is present at the tip of ovule.
- (4) Chalaza lies close to micropylar end of ovule

162. Depicted picture is an enlarged view of microsporangium in which labelled structure A, B, C and D are respectively –



- (1) Tapetum, epidermis, endothecium, middle layer
 - (2) Epidermis, endothecium, tapetum, microspore mother cell
 - (3) Endothecium, middle layer, tapetum, epidermis
 - (4) Endothecium, middle layer, tapetum, microspore mother cell
163. At which temperature, germination of seed can not possible in most of the plants :-
- (1) $10^{\circ} - 15^{\circ}\text{C}$
 - (2) $5^{\circ} - 10^{\circ}\text{C}$
 - (3) $0^{\circ} - 5^{\circ}\text{C}$
 - (4) $20^{\circ} - 25^{\circ}\text{C}$
164. Seed formation without fertilization in flowering plants involves the process of :-
- (1) Sporulation
 - (2) Budding
 - (3) Somatic hybridization
 - (4) Apomixis
165. Which one of the following statements is **not true**?
- (1) Tapetum helps in the dehiscence of anther
 - (2) Exine of pollen grains is made up of sporopollenin
 - (3) Pollen grains of many species cause severe allergies
 - (4) Stored pollen in liquid nitrogen can be used in the crop breeding programmes

166. Which of the following is true regarding pollen viability ?

- (1) Pollen viability is constant in all members of one family
- (2) Pollen viability mainly depends on temperature but not on humidity
- (3) In some members of Solanaceae family pollens remain viable for months
- (4) In wheat and rice pollen remain viable for many months

167. Select the odd one with respect to ploidy level of cell in typical angiosperm :-

- (1) Epicotyl
- (2) Pollen mother cell
- (3) Endosperm
- (4) Embryo

168. Select erroneous (wrong) pair ?

- (1) *Vallisneria* - Pollination by water
- (2) Embryogenesis - Post fertilization event
- (3) Perispermic seed - Black pepper and beet
- (4) Nucellar adventive embryony - Absent in *Citrus*

169. (A) Polar nuclei are situated above the egg apparatus in the large central cell, w.r.t orthotropous ovule.

- (B) Filiform apparatus play important role in guiding the pollen tubes into embryo sac.

- (1) Only (B) is correct
- (2) Both (A) and (B) are incorrect
- (3) (A) and (B) both are correct
- (4) Only (A) is correct

170. **Assertion** :- Polyembryony can be seen in case of *Citrus* and mango seeds.
Reason :- Development of embryo from nucellar cells without fertilisation is found in many varieties of *Citrus* and mango.
- (1) Both Assertion & Reason are True and the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.
171. Non albuminous seed store their reserve food mainly in :-
(1) Cotyledon
(2) Seed coat
(3) Endosperm
(4) Perisperm
172. Pollen tube releases the two male gametes into:-
(1) One male gamete in cytoplasm of synergids and other in cytoplasm of central cell.
(2) Both in cytoplasm of central cell.
(3) Cytoplasm of antipodal cell
(4) Both in cytoplasm of synergid
173. Select the **incorrect** statement from the following-
(1) A typical microsporangium appears nearly tetragonal in outline.
(2) Tapetum nourishes the developing pollen grains.
(3) The anther is four sided structure consisting of four microsporangia.
(4) The cells of the sporogenous tissue undergo meiotic division.
174. Which of the following groups of plants are pollinated by same agents.
(1) Yucca, Salvia, Vallisneria
(2) Hydrilla, Vallisneria, Lotus
(3) Vallisneria, Hydrilla, Zostera
(4) Grass, Yucca, Zostera
175. In the list given below how many structure are diploid ?
Pollen mother cell, Male gamete, Pollen grain, Synergid, Nucellus, Funicle, Perisperm, Integuments.
(1) Five
(2) Six
(3) Four
(4) Seven
176. In which of the following group belong to aeroallergen causing asthmatic problems in human beings ?
(1) Lantana, Grass, Chinrose
(2) Parthenium, Rose, Chenopodium
(3) Mustard, Sorghum, Sunflower
(4) Ambrosia, Amaranthus, Parthenium
177. Select the correct option in which wall layers of anther are correctly arranged from outside to inside :-
(A) Endothelium (B) Epidermis
(C) Middle layers (D) Tapetum
(1) B, C, D, A
(2) B, A, D, C
(3) B, A, C, D
(4) A, B, C, D

178. Read the following statement :-

- (a) Cells of the tapetum are polyploid.
- (b) Cells of tapetum secrete hormones and enzyme.
- (c) Tapetum is inner most layer of anther.
- (d) Tapetum helps in formation of pollen tube.

How many above statements are **correct** ?

- (1) One
- (2) Two
- (3) Three
- (4) Four

179. **Assertion:** Apomixis is a form of asexual reproduction that mimics sexual reproduction.

Reason: In some species, the diploid egg cell is formed without reduction division and develops into the embryo without fertilisation.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

180. Which of the following is example of adventitive embryony ?

- (1) Citrus
- (2) Parthenium
- (3) Ranunculus
- (4) All of these

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